

**Solution**  
**ELECTRICITY -11**  
**Class 10 - Science**  
**Section A**

1. State True or False:

- (i) **(a)** True  
**Explanation:** True
- (ii) **(a)** True  
**Explanation:** True
- (iii) **(a)** True  
**Explanation:** True
- (iv) **(a)** True  
**Explanation:** True
- (v) **(a)** True  
**Explanation:** True
- (vi) **(a)** True  
**Explanation:** True
- (vii) **(a)** True  
**Explanation:** True
- (viii) **(a)** True  
**Explanation:** True
- (ix) **(a)** True  
**Explanation:** True
- (x) **(a)** True  
**Explanation:** True
- (xi) **(b)** False  
**Explanation:** False. An ammeter is a device used for the measurement of current.
- (xii) **(b)** False  
**Explanation:** False. The potential difference is measured by means of an instrument called the voltmeter.
- (xiii) **(a)** True  
**Explanation:** True
- (xiv) **(b)** False  
**Explanation:** False. The voltmeter is always connected in parallel across the points between which the potential difference is to be measured.
- (xv) **(a)** True  
**Explanation:** True
- (xvi) **(a)** True  
**Explanation:** True
- (xvii) **(a)** True  
**Explanation:** True
- (xviii) **(b)** False  
**Explanation:** False. A parallel circuit divides the current through the electrical gadgets.
- (xix) **(a)** True  
**Explanation:** True

**Section B**

2. Fill in the blanks:

- (i) 1. Heating effect
- (ii) 1. Positive
- (iii) 1. Kilowatt-hour
- (iv) 1. Electric circuit
- (v) 1. 1 watt
- (vi) 1. Tungsten
- (vii) 1. Resistivity
- (viii) 1. Rheostat
- (ix) 1. Drawing
- (x) 1. Electric power
- (xi) 1. Insulators
- (xii) 1. Voltmeter
- (xiii) 1. Variable resistance
- (xiv) 1. Parallel
- (xv) 1. Resistor
- (xvi) 1. Ammeter
- (xvii) 1. Higher
- (xviii) 1. Ohm's law
  - 2. Ohm's Law
  - 3. Ohms Law
- (xix) 1. Electric potential
- (xx) 1. Ampere

### Section C

3.

(c) Statement A is false, Statement B is true.

**Explanation:** Electrical resistivity (also known as resistivity) is an intrinsic property of a material that quantifies how strongly a given material opposes the flow of electric current. The low resistivity of a material indicates that the material readily allows the flow of electric current.

Both the resistance and resistivity of a material vary with temperature. The resistivity of a conductor increases with an increase in temperature. The resistivity of semiconductors decreases with an increase in temperature.

4.

(c) Statement A is true; Statement B is false.

**Explanation:**

- Light from a bathroom bulb gets dimmer for a moment when the geyser is switched on. A geyser has a higher power rating and draws a large current from the mains when it is switched on.
- Insulators do not conduct charges; they have high resistivity.

5.

(d) Both A and B are true

**Explanation:** In series combinations, the current in each resistor of the circuit will be the same.

In series combinations, the current simply passes through resistors and do not divide. So the current passing through the different resistors is the same.

Potential Difference is equal to the sum of the potential drop across the individual resistors.

$$\Delta V = IR$$

6.

(d) (A)

**Explanation:** Electrons are negatively-charged particles. The object losing electrons gains a positive charge. The object gaining electrons gains a negative charge.

### Section D

7. (b) A and D  
**Explanation:** B and C are correct. An ammeter is a low resistance device. A semiconductor material has an electrical conductivity value falling between that of a conductor, such as copper, and an insulator, such as glass. As the temperature increases, their resistance decreases.  
 A and D are incorrect. A neutron is a sub-atomic particle that has no charge. One  $\mu A$  is equal to  $10^{-6}$  A. An ampere is a bigger unit. Micro-ampere ( $\mu A$ ) is a smaller unit.
8. (b) (A)  
**Explanation:**  
 As a convention, the direction of flow of positive charge was taken to be the direction of electric current. The direction of flow of electrons is opposite to the direction of conventional current.
9. (a) None of these  
**Explanation:** According to Joule's law of heating,  $H = I^2 R t$  where the heat produced in a resistor is directly proportional to the square of the current for a given resistance, directly proportional to the resistance for a given current and directly proportional to the time for which the current flows through the resistor.
10. (a) Both wires are made of same material and are at same temperature.  
**Explanation:** Resistance (R) of a conductor at a given temperature:
  - is directly proportional to the length (l)
  - is inversely proportional to the area of cross-section (a)
  - depends on the nature of the material ( $\rho$ ) $R = \rho \frac{l}{A}$  where  $\rho$  is a constant of proportionality and is called the electrical resistivity of the material of the conductor.  
 So, the resistance of two wires of the same length and the same area of the cross-section will be the same if both the wires have the same resistivity ( $\rho$ ) i.e. they are made up of the same material and are at the same temperature. Both resistance and resistivity of a material vary with temperature. The resistivity of a conductor increases with the increase in temperature.
11. (c) A and C  
**Explanation:** Resistance (R) of a conductor at a given temperature:
  - R is directly proportional to the length (l).
  - R is inversely proportional to the area of cross-section (a).
  - R depends on the nature of the material ( $\rho$ ). $R = \rho \times \frac{l}{a}$
12. (b) An ammeter is connected in series in a circuit and a voltmeter is connected in parallel  
**Explanation:** An ammeter is connected in series in a circuit and a voltmeter is connected in parallel

### Section E

13. (d) Ammeter of range 0-5 A and voltmeter of range 0-10 V  
**Explanation:** Voltmeter has to read beyond 6 V and with 0-2A range, ammeter will have a better least count but net current drawn for 6V is 3A.
14. (b) ammeter  $A_2$  and voltmeter  $V_2$   
**Explanation:**  $I = \frac{V}{R} = \frac{6}{\frac{4 \times 12}{16}} = \frac{6 \times 16}{4 \times 12} = 2A$   
 So  $V_2$  of range 0 to 5V and  $A_2$  of range 0 to 3 A range is suited for measuring upto 2A.
15. (d) one terminal of each of the two resistors and these terminals are not interconnected.

**Explanation:** One terminal of each resistor should be connected and the voltmeter has to be connected between the other free terminals of both the resistors.

16.

(c) A and E

**Explanation:** The key has to be removed once the reading is taken otherwise temperature will rise due to the passage of current. Copper wire cannot be as thin as possible because it will have more resistance.

17. (a)  $R_S > R_P$

**Explanation:** In series  $R_S = R_1 + R_2$ ; In parallel,  $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ .

18.

(b) ammeter  $A_2$  and voltmeter  $V_2$

**Explanation:** The overall range of the voltage is from 0 to 1.5 V and that of current is from 0 to  $\frac{1.5}{3}$  A = 0.5 A. We, therefore, prefer, instruments that cover these ranges and also have a better least count.

19.

(c) X, Y and Z

**Explanation:** The current in the series does not depend upon order in which these instruments are connected. Where they are connected does not matter. They should be in series for the flow to be observed.

20.

(b) Milliammeter  $A_1$  and voltmeter  $V_1$

**Explanation:** Milliammeter  $A_1$  and voltmeter  $V_1$  gives maximum measuring range with lowest least count. So, the combination of these two is best for two resistances connected in series.

### Section F

21. a. When an electric current passes through a conductor (like a high resistance wire) the conductor becomes hot after some time and produces heat. This is called heating effect of Electric Current.

**Ex:** A bulb becomes hot after its use for some time. This is because of heating effect of electric current.

b. We Know that

$$H = P \times t$$

$$H = \frac{\text{Energy}}{\text{Time}} \times t$$

$$\therefore \text{Energy} = V.Q$$

$$H = \frac{V.Q.t}{t}$$

$$H = Vit$$

$$\therefore V = IR$$

$$H = I^2 R t$$

Other form

$$H = V \left[ \frac{V}{R} \right] t$$

Where

I = Current

R = Resistance

t = Time

c. Two devices which work on the heating effect of electric current are:

i. Electric heater

ii. Electric iron

22. a. In first case :  $1 = 1\Omega$  and  $2 = 10^6\Omega$

$$\therefore \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{1} + \frac{1}{10^6} = \frac{10^6 + 1}{10^6}$$

$$\frac{1}{R_p} = \frac{1,000,000 + 1}{1,000,000} = \frac{1,000,001}{1,000,000}$$

$$R_p = \frac{1,000,000}{1,000,001} = 0.999999\Omega$$

$$\text{or } R_p = 1\Omega$$

b. In second case,  $R_1 = 1\Omega$ ,  $R_2 = 10^3\Omega$ ,  $R_3 = 10^6\Omega$

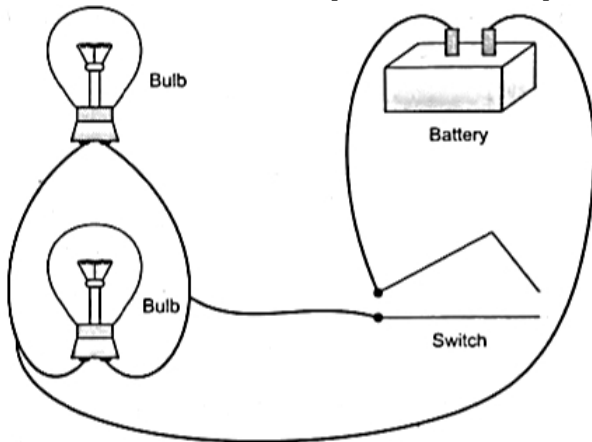
$$\frac{1}{R_p} = \frac{1}{1} + \frac{1}{10^3} + \frac{1}{10^6} = \frac{10^5 + 10^3 + 1}{10^5}$$

$$\frac{1}{R_p} = \frac{1000000 + 1000 + 1}{1000000} = \frac{1001001}{1000000}$$

$$R_p = \frac{1,000,000}{1,001,001} = 0.999000001\Omega$$

$$R_p = 1\Omega$$

23. The two bulbs are connected in parallel and the complete circuit diagram is drawn below:



The reason for connecting the two bulbs in parallel is that (i) both the bulbs glow at the same voltage, and (ii) if one bulb stops glowing, the other bulb remains unaffected.

24. Given :-

Resistor,  $R_1 = 4\Omega$

Resistor,  $R_2 = 8\Omega$

Resistor,  $R_3 = 8\Omega$

Potential Difference,  $V = 8$  volts.

i. The potential difference across  $4\Omega$  resistors,  $V = IR = 1 \times 4 = 4V$

ii. Power dissipated in  $4\Omega$  resistors,  $P = I^2R = (1A)^2(4\Omega) = 4W$

25.  $\therefore$  The resistance of a conductor is directly proportional to the length of the conductor ( $R \propto l$ ).

$\therefore$  The resistance of the metallic wire, when it is cut into ten parts of equal length,  $r = \frac{R}{10}$

Two such pieces when joined in series, the equivalent resistance of these two parts =  $r + r$

$$= 2r = \frac{2R}{10}$$

$\therefore$  Equivalent resistance of two parts

$$= 2 \times \frac{R}{10}$$

$$= \frac{R}{5}$$

5 such parts are connected in parallel.

$\therefore$  the total resistance  $R'$ ,

$$\frac{1}{R'} = \frac{1}{\frac{R}{5}} + \frac{1}{\frac{R}{5}} + \frac{1}{\frac{R}{5}} + \frac{1}{\frac{R}{5}} + \frac{1}{\frac{R}{5}}$$

$$= \frac{25}{R}$$

$$\text{Hence, } R' = \frac{R}{25}$$

26.  $R_1$  and  $R_2$  are in series,

$$R_s = R_1 + R_2 = 3 + 3 = 6\Omega$$

$R_s$  and  $R_3$  are in parallel.

$$\therefore \frac{1}{R_p} = \frac{1}{R_s} + \frac{1}{R_3}$$

$$= \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

$$\Rightarrow R_p = 2\Omega$$

$R_4$ ,  $R_p$ ,  $R_5$  are in series,

$$\therefore R_{s'} = R_4 + R_p + R_5 = 0.5 + 2 + 0.5 = 3\Omega$$

$$\text{Current, } I = \frac{V}{R_{s'}} = \frac{3}{3} = 1A$$

27. i.  $R_{AB} = R_1 + R_2 + R_3$

$$\text{II. } \frac{1}{R_{AB}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

- ii. According to ohm's law ( $V = IR$ ). Slope of V-I graph gives resistance. So greater the slope of graph, greater will be its resistance. Since we know that equivalent resistance in series is greater than the equivalent resistance in Parallel. So the first (I) graph is correct representation of V-I graph in series and Parallel combination.

$$28. P_1 = \frac{1000W}{1000} = 1kW$$

$$t_1 = 5 \text{ h}$$

$$P_2 = \frac{400W}{1000} = 0.4 \text{ kW}$$

$$t_2 = 10 \text{ h}$$

No. of days  $n = 30$

Energy consumed by heater:

$$E_1 = P_1 \times t_1 \times n = 1 \text{ kW} \times 5 \text{ h} \times 30 = 150 \text{ kWh}$$

Energy consumed by refrigerator:

$$E_2 = P_2 \times t_2 \times n = 0.4 \text{ kW} \times 10 \text{ h} \times 30 = 120 \text{ kWh}$$

$$\text{Total energy} = (150 + 120) \text{ kWh} = 270 \text{ kWh}$$

We know that  $1\text{kWh} = 1\text{unit}$ , so

$$270 \text{ kWh} = 270 \text{ units}$$

Cost of 1 unit is Rs. 6.00

$$\text{Total cost} = 270 \times 6 = \text{Rs. } 1620$$

29. i. Let the equivalent resistance of parallel resistors  $R_1$  and  $R_2 = R'$ . The equivalent resistance of parallel resistors  $R_3$ ,  $R_4$  and  $R_5 = R''$ .

Then, we have

$$\Rightarrow \frac{1}{R'} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R'} = \frac{1}{10} + \frac{1}{40}$$

$$\Rightarrow \frac{1}{R'} = \frac{4+1}{40}$$

$$\Rightarrow R' = \frac{40}{5} = 8\Omega$$

Similarly,

$$\frac{1}{R''} = \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5}$$

$$\Rightarrow \frac{1}{R''} = \frac{1}{30} + \frac{1}{20} + \frac{1}{60}$$

$$\Rightarrow R'' = \frac{60}{6} = 10\Omega$$

$$R' = 8\Omega$$

$$R'' = 10\Omega$$

As  $R'$  and  $R''$  are connected in series.

Thus, the total resistance in the circuit,

$$R = R' + R'' = 8\Omega + 10\Omega = 18\Omega$$

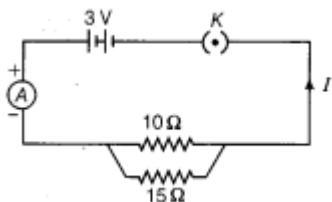
- ii. According to Ohm's law

$$I = \frac{V}{R}$$

$$= \frac{12V}{18\Omega}$$

$$= 0.67 \text{ A}$$

30. Given,



- i. Current through

$$\text{a. } 10\Omega \text{ resistor, } I_1 = \frac{V}{R} = \frac{3}{10} = 0.3A$$

$$\text{b. } 15\Omega \text{ resistor, } I_2 = \frac{V}{R} = \frac{3}{15} = 0.2A$$

ii. Current flowing through the circuit =  $0.3 + 0.2 = 0.5\text{ A}$

Thus, the ammeter reading is  $0.5\text{ A}$ .

31. a. (i) Length of the conductor ( $l$ )

(ii) Area of cross-section of the conductor ( $A$ )

b. Radius of wire,  $r = 0.01\text{ cm} = 0.01 \times 10^{-2}\text{ m}$

Resistance,  $R = 10\ \Omega$

Resistivity,  $\rho = 50 \times 10^{-8}\ \Omega\text{m}$

$$R = \rho \frac{l}{A} = \rho \frac{l}{\pi r^2} \Rightarrow l = \frac{R \pi r^2}{\rho}$$

$$l = \frac{10\ \Omega \times 22 \times (0.01 \times 10^{-2})^2\text{ m}}{7 \times 50 \times 10^{-8}\ \Omega\text{m}}$$

$$= \frac{22}{35}\text{ m} = 0.629\text{ m} = 0.628\text{ m} = 0.62\text{ m}$$

32. i.



If we connected two  $9\text{ ohm}$  resistors in parallel and other  $9\text{ ohm}$  resistor in series as shown in figure:

$$\frac{1}{R_p} = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}$$

$$R_p = \frac{9}{2}\ \Omega$$

$$R = 9\ \Omega + \frac{9}{2}\ \Omega = 13.5\ \Omega$$

ii.



If we connected two  $9\text{ ohm}$  resistors in series and other  $9\text{ ohm}$  resistor in parallel as shown in figure:

$$R_s = 9\ \Omega + 9\ \Omega = 18\ \Omega$$

$$\frac{1}{R} = \frac{1}{18} + \frac{1}{9} = \frac{3}{18}$$

$$\therefore R = 6\ \Omega$$

33. i. Here  $I = 2.5\text{ mA} = 2.5 \times 10^{-3}\text{ A}$ ;  $V = 12\text{ volt}$ ;  $R = ?$

$$V = IR \text{ or } R = \frac{V}{I}$$

$$R = \frac{12}{2.5 \times 10^{-3}} = \frac{12}{25 \times 10^{-4}} \times \frac{120000}{25} = 4800\ \Omega$$

ii. According to the question,  $H = 320\text{ J}$ ,  $t = 10\text{ s}$ ,  $R = 2\ \Omega$ ,  $I = ?$

We know that,  $H = I^2 R t$

$$\Rightarrow I = \sqrt{\frac{H}{Rt}}$$

$$= \sqrt{\frac{320}{2 \times 10}}$$

$$= 4\text{ A}$$

Thus, the amount of current flowing through the resistor is  $4\text{ A}$ .

34. i.
- When the key is plugged-in, current starts in coil-1, the magnetic field around the coil is changed. This produces induced current in the coil-2 and galvanometer shows deflection
  - There is no change in magnetic field when a steady current starts flowing in the circuit.

ii. Galvanometer shows deflection in the opposite direction.

iii. Conclusion: Induced current is produced only when there is a change in magnetic field which occurs only when the key is plugged in or plugged out.

35. **Ohm's law:** It states that the current flowing through a conductor between two points is directly proportional to the potential difference across the two points.

The arrangement of components in a circuit is normally from the battery to the Ammeter then to the resistor. We have been given the symbols for the Resistor, Ammeter and other components.

The correct order is therefore given as:

B, A, K, R, B

We, therefore conclude that student Y place them in the right order.

36. i.  $P = VI$

$$\text{Then, } I = \frac{P}{V}$$

When heating is at the maximum rate,

$$I = \frac{840 \text{ W}}{220 \text{ V}} = 3.82 \text{ A}$$

and the resistance the electric iron is

$$R = \frac{V}{I} = \frac{220 \text{ V}}{3.82 \text{ A}} = 57.59 \Omega$$

when heating is at the minimum rate,

$$I = \frac{360 \text{ W}}{220 \text{ V}} = 1.64 \text{ A}$$

and the resistance of the electric iron is

$$R = \frac{V}{I} = \frac{220 \text{ V}}{1.64 \text{ A}} = 134.15 \Omega$$

ii. Energy consumed  $E_1 = P_1 t_1 = 250 \text{ W} \times 1 \text{ h} = 250 \frac{\text{J}}{\text{s}} \times 3600 \text{ s}$

$$E_1 = 900000 \text{ J}$$

$$E_2 = P_2 t_2 = 12000 \text{ W} \times 10 \text{ min}$$

$$E_2 = 1200 \frac{\text{J}}{\text{s}} \times 600 \text{ s}$$

$$E_2 = 720000 \text{ J}$$

$\therefore 250 \Omega$  TV set consumes more energy.

37. The resistance of electric lamp,  $R_1 = 20\Omega$ ,

The resistance of the conductor connected in series,  $R_2 = 4\Omega$

Then the total resistance in the circuit

$$R = R_1$$

$$+ R_2$$

$$R_s = 20\Omega + 4\Omega = 24\Omega$$

The total potential difference across the two terminals of the battery

$$V = 6 \text{ V.}$$

Now by Ohm's law, the current through the circuit is given by

$$I = \frac{V}{R}$$

$$\frac{6 \text{ V}}{24\Omega}$$

$$= 0.25 \text{ A}$$

Applying Ohm's law to the electric lamp and conductor separately,

We get potential difference across the electric lamp,

$$V_1 = 20\Omega \times 0.25 \text{ A}$$

$$= 5 \text{ V}$$

and,

$$\text{that across the conductor, } V_2 = 4\Omega \times 0.25 \text{ A}$$

$$= 1 \text{ V}$$

and,

$$\text{that across the conductor, } V_2 = 4\Omega \times 0.25 \text{ A}$$

$$= 1 \text{ V}$$

Suppose that we like to replace the series combination of electric lamp and conductor by a single and equivalent resistor. Its resistance must be such that a potential difference of 6 V across the battery terminals will cause a current of 0.25 A in the circuit.

The resistance R of this equivalent resistor would be

$$R = \frac{V}{I}$$

$$= \frac{6 \text{ V}}{0.25 \text{ A}}$$

$$= 24\Omega$$

This is the total resistance of the series circuit; it is equal to the sum of the two resistances.



38. i. It is given that potential difference (V) = 220 V.

Resistance of coil A ( $R_A$ ) = Resistance of coil B ( $R_B$ ) =  $24\Omega$

When either coil A or B is used separately, the current ( $I$ ) =  $\frac{V}{R} = \frac{220\text{ V}}{24\Omega} = 9.2\text{ A}$

When two coils are used in series, total resistance,  $R_S = R_A + R_B$

$$= R_A + R_B = 24 + 24 = 48\Omega$$

Current flowing ( $I$ ) =  $\frac{V}{R_S} = \frac{220\text{ V}}{48\Omega} = 4.6\text{ A}$

When the two coils are joined in parallel, total resistance ( $R_p$ ) =  $\frac{1}{24} + \frac{1}{24} = \frac{2}{24}$

ii.  $I = 10\text{ A}$ ,  $V = 220\text{ V}$

$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{220}{10} \\ &= 22\Omega \end{aligned}$$